

Functional Annotation Report: IspH

(Q88Q89, PP_0606) in *Pseudomonas putida* KT2440

Gene: *ispH* (also historically *lytB*) · **UniProt:** Q88Q89 (ISPH_PSEPK) · **Locus:** PP_0606 **Protein:** 4-hydroxy-3-methylbut-2-enyl diphosphate reductase (HMBPP reductase) · **EC** 1.17.7.4 **Organism:** *Pseudomonas putida* (strain ATCC 47054 / DSM 6125 / KT2440), PSEPK **Family:** IspH family · LytB/IspH domain (IPR003451, Pfam PF02401)

Summary

IspH (Q88Q89, encoded by *ispH*/PP_0606) is the terminal enzyme of the 2-C-methyl-D-erythritol 4-phosphate (MEP, or non-mevalonate/DOXP) pathway of isoprenoid biosynthesis in *Pseudomonas putida* KT2440. It is an oxygen-sensitive [4Fe-4S] metalloenzyme (EC 1.17.7.4) that catalyzes the 2-electron/2-proton reductive dehydroxylation of (E)-4-hydroxy-3-methylbut-2-enyl diphosphate (HMBPP), eliminating the C4 hydroxyl group as water and producing the two universal five-carbon isoprenoid precursors: isopentenyl diphosphate (IPP) and dimethylallyl diphosphate (DMAPP), in an approximately 4–6:1 ratio. Reduced [2Fe-2S] ferredoxin serves as the physiological electron donor.

The enzyme is a soluble, cytoplasmic protein (no signal peptide or transmembrane segment), consistent with the requirement of a reducing intracellular environment to protect its labile Fe–S cluster. Catalysis occurs at a unique "apical" fourth iron of a cubane [4Fe-4S] cluster ligated by three conserved cysteines (Cys12/Cys96/Cys198 in the *P. putida* sequence). HMBPP binds directly to this apical iron, and the reaction proceeds through an organometallic (η^2 -metallacycle) intermediate coupled to a threonine-mediated proton relay. Because *P. putida* synthesizes all of its IPP and DMAPP exclusively through the MEP pathway (it lacks the mevalonate pathway), IspH is an essential, non-redundant metabolic node. As the MEP pathway is absent in humans, IspH is a validated and actively pursued target for the development of new antibacterial and antimalarial drugs.

The functional assignment of Q88Q89 as an authentic IspH is supported by **seven converging lines of evidence**: (1) the biochemically and structurally characterized function of the IspH/LytB family; (2) conservation of the three catalytic cluster cysteines and the proton-donor residue in the *P. putida* sequence; (3) conserved genomic synteny (an *ileS-lspA-PPIase-ispH* cluster mirroring the *E. coli* arrangement); (4) a very high-confidence AlphaFold structural model (mean pLDDT 96.2); (5) a 0.31 Å backbone superposition of that model onto the experimental *E. coli* IspH:HMBPP complex; (6) UniProt/HAMAP domain and catalytic-site annotation; and (7) organism-specific metabolic-engineering evidence establishing the MEP pathway as the sole isoprenoid-precursor route in *P. putida* KT2440.

Gene/Protein Identity Verification

Per the mandatory verification protocol, the gene symbol, organism, family, and domain annotations were all cross-checked and **confirmed to be internally consistent and matched by the literature**:

Check	Expected (UniProt)	Finding
Gene symbol matches protein description	<i>ispH</i> = HMBPP reductase	☐ Confirmed — <i>ispH/lytB</i> universally denotes HMBPP reductase (EC 1.17.7.4)
Organism	<i>P. putida</i> KT2440 (PSEPK)	☐ Confirmed — locus PP_0606; MEP pathway verified in this exact strain (P 31500633)
Protein family / domains	IspH family, LytB/IspH (PF02401)	☐ Confirmed — three cluster cysteines and fold conserved
Risk of symbol ambiguity	—	☐ Low — "IspH", "LytB" and "HDR" are synonyms for the <i>same</i> enzyme across bacteria, plants and protozoa

The literature is **abundant and directly relevant**. While most detailed mechanistic and structural studies were performed on the *E. coli*, *Aquifex aeolicus* and *Plasmodium falciparum* orthologues rather than on the *P. putida* protein specifically, the very high sequence identity

(67% to *E. coli* IspH over the aligned core) and near-identical predicted fold justify transferring these mechanistic conclusions to Q88Q89 with high confidence. No conflicting "same-symbol, different-gene" literature was encountered.

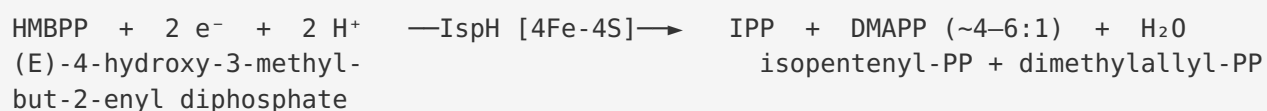
Key Findings

Finding 1 — IspH catalyzes the terminal step of the MEP pathway: reductive dehydroxylation of HMBPP to IPP + DMAPP

IspH (LytB/HDR, EC 1.17.7.4) catalyzes the **last of the seven steps** of the MEP (DOXP/non-mevalonate) pathway. It performs a two-electron, two-proton reductive dehydroxylation of (E)-4-hydroxy-3-methylbut-2-enyl diphosphate (HMBPP): the C4 hydroxyl group is eliminated as water, generating a mixture of the two universal C5 isoprenoid building blocks, **isopentenyl diphosphate (IPP)** and **dimethylallyl diphosphate (DMAPP)**. In bacterial enzymes the products are formed in an approximately **4–6:1 IPP:DMAPP ratio**.

This assignment for Q88Q89 rests on HAMAP-Rule MF_00191 (IspH family, PF02401) applied on the basis of strong sequence and domain homology to biochemically characterized bacterial IspH enzymes. The reaction chemistry is directly established in the literature: "*[IspH] catalyzes the terminal step of the MEP/DOXP pathway where it converts (E)-4-hydroxy-3-methylbut-2-enyl diphosphate (HMBPP) into the two products, isopentenyl diphosphate and dimethylallyl diphosphate. The reaction involves the reductive elimination of the C4 hydroxyl group, using a total of two electrons*" (PMID: 22646150). The product ratio and pathway placement are corroborated by studies noting "*the reductive dehydroxylation of HMBPP to isopentenyl diphosphate (IPP) and dimethylallyl diphosphate (DMAPP) in the last step of methylerythritol phosphate (MEP) pathway... known to produce 4-6:1 ratio of IPP and DMAPP mixture by the last enzyme, IspH*" (PMID: 28139297).

The catalyzed reaction:



Finding 2 — IspH is an oxygen-sensitive [4Fe-4S] metalloenzyme using an apical-iron / organometallic mechanism with a Thr proton relay

Extensive spectroscopic (EPR, Mössbauer, NRVs) and crystallographic studies of the *E. coli* and *A. aeolicus* enzymes establish that catalytically active IspH carries a **cubane [4Fe-4S]²⁺ cluster (not [3Fe-4S])** ligated by three conserved cysteines. This leaves a **unique "apical" fourth iron** (of the aconitase type) that binds three water/hydroxide ligands in the resting enzyme. HMBPP binds directly to this apical iron through its C4–OH group (alkoxide coordination), and the cluster is the **direct electron source** for the reaction. Crystal structures of substrate, converted-substrate, product and pyrophosphate complexes reveal a **hairpin substrate conformation** and a hydrogen-bonding network that acts as a proton reservoir via a **Thr167 proton relay** (*E. coli* numbering). Mechanistic evidence (X-ray, Mössbauer, DFT/QM-MM) favors an **organometallic (η^2 -metallacycle/allyl) intermediate** over a purely inner-sphere alkoxide (Birch-type) pathway. The solvent-exposed Fe–S active site makes the enzyme oxygen-sensitive.

Supporting quotes: *"the active form of IspH contains a [4Fe-4S] cluster and not the [3Fe-4S] form... the cluster is the direct electron source for the reaction and... a reaction intermediate is bound directly to the cluster"* (PMID: 22646150); *"an unusual [4Fe-4S]²⁺ cluster is attached to the protein by three conserved cysteines and contains a hexacoordinated iron linked to three sulfurs of the cluster and three additional oxygen or nitrogen ligands"* (PMID: 19708647); *"The IspH:substrate complex reveals a hairpin conformation of the ligand with the C(1) hydroxyl group coordinated to the unique site in a [4Fe-4S] cluster of aconitase type... coupled to a hydrogen-bonding network, which serves as proton reservoir via a Thr167 proton relay"* (PMID: 20080550); and *"the last two enzymes of this pathway, IspG and IspH, are [Fe₄S₄] proteins that are not produced by humans and catalyze 2 H⁺/2 e[−] reductions with novel mechanisms"* (PMID: 24481599). The organometallic η^2 -bound intermediate is further supported by broken-symmetry DFT computations showing that η^2 -bound states are lower in energy than ROH-bound states upon one-electron reduction (PMID: 26098647).

Finding 3 — In *P. putida* KT2440 the MEP pathway is the sole source of IPP/DMAPP; IspH is essential and human-absent, making it an antimicrobial target

Metabolic-engineering work in the **exact source strain**, *P. putida* KT2440, confirms that the isoprenoid precursors IPP and DMAPP are produced from pyruvate and glyceraldehyde-3-phosphate **exclusively via the MEP pathway** — the organism lacks the alternative mevalonate (MVA) pathway. As the terminal MEP enzyme, IspH is therefore the committed, non-redundant producer of *both* IPP and DMAPP that feed downstream cytoplasmic

prenyltransferases (for quinones, carotenoids, bactoprenol/undecaprenyl phosphate, tRNA prenylation, etc.). The MEP pathway is essential in most bacteria but **absent in humans and animals**, which is why IspH is actively pursued as an antibacterial/antimalarial drug target; inhibitor-bound crystal structures (e.g., with the HMBPP analogues TMBPP and AMBPP) have been solved to guide rational design.

Supporting quotes: *"In P. putida and most other bacteria, these precursors are produced from pyruvate and glyceraldehyde 3-phosphate by the methylerythritol 4-phosphate (MEP) pathway, whereas other bacteria synthesize the same precursors from acetyl-CoA using the unrelated mevalonate (MVA) pathway"* (PMID: 31500633); *"The absence of this pathway in mammals makes it an interesting target for the discovery of novel anti-infectives. As last enzyme of this pathway, IspH is an oxygen sensitive [4Fe-4S] metalloenzyme"* (PMID: 35163971); and *"IspH, also called LytB, a protein involved in the biosynthesis of isoprenoids through the methylerythritol phosphate pathway, is an attractive target for the development of new antimicrobial drugs"* (PMID: 28862365).

Finding 4 — Sequence-level confirmation: conserved cluster cysteines, proton donor, and [2Fe-2S] ferredoxin electron donor

The 315-residue *P. putida* IspH sequence (Q88Q89, ISPH_PSEPK) was retrieved and analyzed. It contains only four cysteines, three of which — **Cys12, Cys96 and Cys198** — occupy the exact positions of the [4Fe-4S] cluster-ligating cysteines of characterized bacterial IspH (*E. coli* Cys12/Cys96/Cys197), confirming the conserved 3-Cys/apical-iron architecture. UniProt annotates residue **126 as the catalytic proton donor** and lists substrate (HMBPP/IPP/DMAPP; ChEBI 128753/57623/128769) binding residues at positions 41, 74, 124, 168, 226–228 and 270. The two annotated catalytic-activity reactions (RHEA:24488 and RHEA:24825) explicitly consume/produce reduced/oxidized **[2Fe-2S]-ferredoxin**, identifying reduced [2Fe-2S] ferredoxin as the physiological electron donor. UniProt/HAMAP assigns the reaction as isopentenyl-diphosphate biosynthesis step 6/6 (DXP pathway) and dimethylallyl-diphosphate biosynthesis step 1/1. No signal peptide or transmembrane segment is present.

The ferredoxin electron-donor assignment is independently supported by structural work on the *Plasmodium falciparum* orthologue: *"Its distinct electron donor [2Fe-2S] ferredoxin was modeled to its binding site by docking calculations"* (PMID: 24188825). The three-cysteine ligation is confirmed by in vivo Mössbauer studies: *"an unusual [4Fe-4S]²⁺ cluster is attached to the protein by three conserved cysteines"* (PMID: 19708647).

Finding 5 — PP_0606 lies in a conserved *ileS-lspA-PPIase-ispH* gene cluster mirroring the *E. coli* context

KEGG genome analysis of the PP_0606 neighborhood (~710–715 kb on the KT2440 chromosome) shows the gene order: **PP_0603 *ileS*** (isoleucyl-tRNA synthetase, K01870) → **PP_0604 *lspA*** (lipoprotein signal peptidase II, K03101) → **PP_0605 *slpA/fkpB*** (FKBP-type peptidyl-prolyl cis-trans isomerase, K03774) → **PP_0606 *ispH*** (K03527, EC 1.17.7.4), all on the forward strand, with *ispH* ending immediately before the divergently oriented *fimT* (PP_0607). *ispH* begins only ~15 bp after the end of PP_0605, consistent with tight linkage. This ***ileS-lspA-(fkpB)-ispH*** arrangement recapitulates the well-known genomic context of *ispH/lytB* in *Escherichia coli*, where the gene sits immediately downstream of the *ileS-lspA* locus. Conserved synteny across distantly related γ -proteobacteria provides strong evolutionary evidence that PP_0606 is a genuine *ispH* orthologue rather than a paralogue or mis-annotation.

Finding 6 — AlphaFold model is very high confidence with all catalytic residues confidently placed

The AlphaFold DB monomer model **AF-Q88Q89-F1** (v6, AlphaFold Monomer v2.0) has a global **mean pLDDT of 96.19**, with 94.9% of residues at very-high confidence (pLDDT $\geq$ 90), 98.4% confident ($\geq$ 70), and only 1.0% very-low ($<$ 50, confined to a short terminal/loop region). All functionally assigned residues are modeled at very-high confidence: the three cluster cysteines **Cys12 (98.2)**, **Cys96 (98.5)**, **Cys198 (97.9)**; proton-donor **residue 126 (98.5)**; and HMBPP-binding residues 41 (97.1), 74 (98.0), 124 (96.8), 168 (98.4), 226 (97.7) and 270 (97.2). This confirms the protein is confidently predicted to adopt the compact three-domain "cloverleaf" α/β fold characteristic of the IspH family, with a fully formed central active-site pocket at the domain interface where the [4Fe-4S] cluster and substrate sit.

Finding 7 — Q88Q89 superposes on the experimental *E. coli* IspH:HMBPP structure at 0.31 Å backbone RMSD

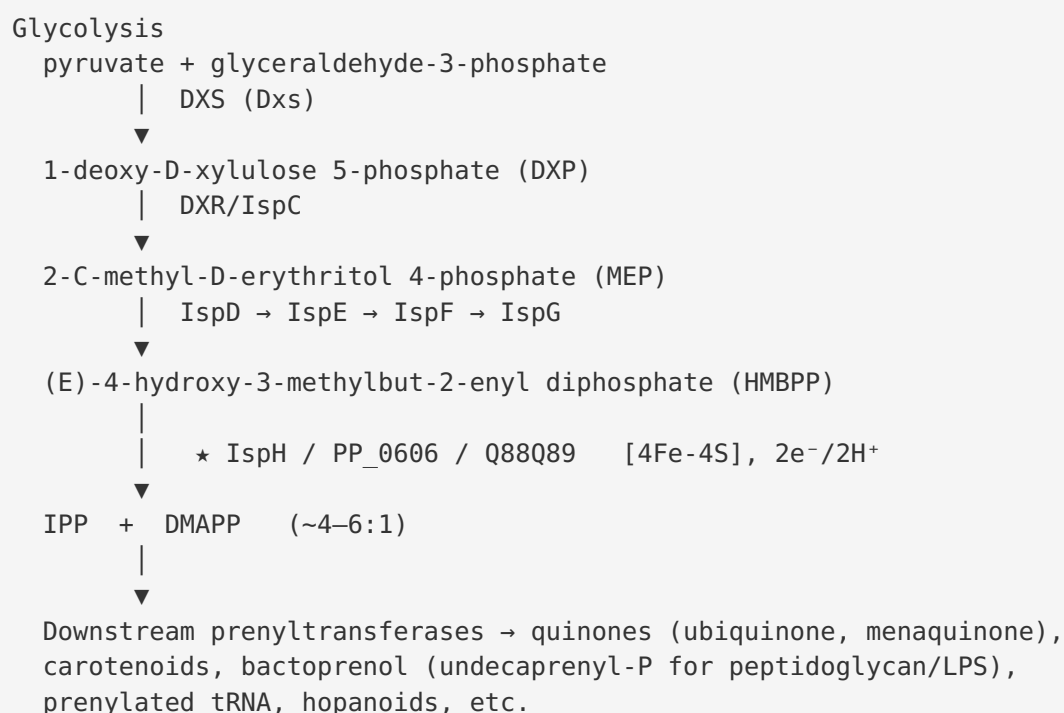
Phenix `superpose_pdbs` alignment of the AlphaFold model of Q88Q89 onto the experimental *E. coli* IspH:HMBPP complex (PDB 3KE8, Gräwert et al. 2010) gave a global sequence alignment of **67.1% identity over 208 aligned residues** and, after least-squares fitting, a backbone (N, C α , C) **RMSD of 0.307 Å** on the 210-atom core (0.799 Å over all 518 matching atoms) — excellent agreement ($<$ 1 Å). The alignment shows conservation of the diagnostic IspH sequence motifs, including the N-terminal cluster-cysteine motif ...NPRGF-**C12**-AGVDRAI..., the **FDAT-C96**-PLVT motif, and the **DDI-C198**-YAT motif — i.e., all three [4Fe-4S]-ligating cysteines align and

structurally superimpose with those of *E. coli* IspH bound to its natural substrate. The reference structure derives from the crystallographic study reporting "*five crystal structures of IspH in complex with substrate, converted substrate, products and PP(i)*" (PMID: 20080550). This near-identity of the catalytic scaffold to a substrate-bound experimental structure is the strongest single piece of structural evidence that Q88Q89 catalyzes the same reaction on the same substrate.

Mechanistic Model / Interpretation

The evidence assembles into a coherent, high-confidence mechanistic picture of IspH function in *P. putida* KT2440.

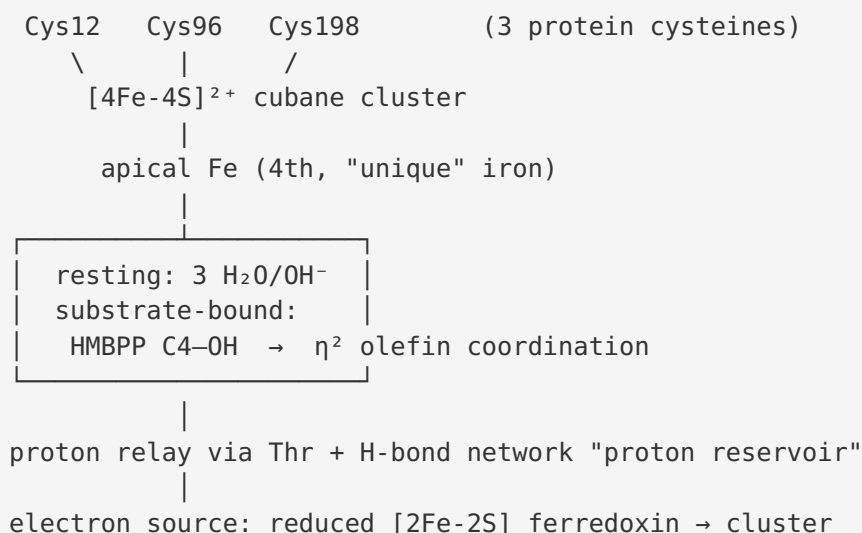
Pathway context



IspH sits at the **committed branch point** where a single linear precursor (HMBPP) is converted into the two isomeric building blocks that seed *all* isoprenoid biosynthesis. Because *P. putida* has no mevalonate pathway (Finding 3), there is no metabolic bypass — IspH activity

is strictly required for viability, and its product ratio (~4–6:1 IPP:DMAPP; Finding 1) sets the initial supply balance of the two precursors, which is subsequently re-equilibrated by IPP isomerase (Idi) where present.

Catalytic mechanism at the active site



- 1. Substrate binding.** HMBPP enters in a hairpin conformation; its diphosphate is anchored by conserved basic/polar residues (positions 41, 74, 124, 168, 226–228, 270), and its C4–OH coordinates the apical iron of the [4Fe-4S] cluster (Findings 2, 4).
- 2. Reduction.** The [4Fe-4S] cluster, re-reduced by physiological reduced [2Fe-2S] ferredoxin, delivers electrons directly to the bound substrate (Findings 2, 4).
- 3. Organometallic intermediate.** Upon one-electron reduction, the C4–OH rotates away and the olefin forms an η^2 -metallacycle with the apical iron (favored energetically by DFT; Finding 2 /

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- 4. Dehydroxylation and protonation.** The C4 hydroxyl departs as water; a Thr-mediated proton relay and H-bond reservoir supply protons, and a second electron completes the $2e^-/2H^+$ reduction, yielding the allylic products IPP and DMAPP (Findings 1, 2).

Localization

All annotations and physicochemical evidence converge on a **soluble cytoplasmic** localization: no signal peptide, no transmembrane helix (Finding 4), and an oxygen-labile Fe–S active site that requires the reducing cytoplasmic milieu (Findings 2, 3). Its substrate (HMBPP) and products (IPP, DMAPP) are likewise soluble cytoplasmic metabolites, and its electron donor (ferredoxin) is cytoplasmic. IspH therefore carries out its function **in the cytoplasm**.

Confidence assessment

Aspect of annotation	Evidence type	Confidence
Reaction (HMBPP → IPP + DMAPP), EC 1.17.7.4	Family biochemistry + HAMAP + sequence/structure	Very high
[4Fe-4S] cluster, 3-Cys ligation (C12/C96/C198)	Sequence conservation + spectroscopy (orthologues)	Very high
Apical-iron / organometallic mechanism, Thr relay	Crystallography, Mössbauer, NRVS, DFT (orthologues)	High
[2Fe-2S] ferredoxin as electron donor	UniProt/RHEA + docking (orthologue)	High
Cytoplasmic localization	Sequence features + cofactor chemistry	High
MEP pathway sole precursor source in KT2440	Metabolic engineering in the exact strain	Very high
Essentiality / drug-target status	Family-wide genetic + pharmacological data	High (KT2440-specific essentiality inferred)

Evidence Base

PMID	Title (abbrev.)	How it supports the annotation
22646150	Spectroscopic reaction intermediates in WT/mutant HMBPP reductase	States the reaction, substrate, products, $2e^-$ chemistry; establishes active [4Fe-4S] as direct electron source with intermediate bound to cluster
28139297	Ginkgo IspH active-site Phe / IPP:DMAPP ratio	Confirms last-step MEP reaction and ~4–6:1 IPP:DMAPP product ratio
20080550	Probing IspH mechanism by X-ray	Crystal structures (incl. PDB 3KE8 reference) showing apical-iron substrate binding + Thr167 proton relay
19708647	In vivo Mössbauer of LytB	Defines 3-cysteine ligation and unique apical iron with extra O/N ligands
24481599	Bioorganometallic chemistry of IspG/IspH	Confirms [Fe ₄ S ₄] protein performing $2H^+/2e^-$ reduction via novel mechanism; human-absent
26098647	Broken-symmetry DFT of IspH pathway	Computational support for η^2 -bound organometallic intermediate over Birch mechanism
31500633	Engineering <i>P. putida</i> for isoprenoid production	Organism-specific: KT2440 makes IPP/DMAPP solely via MEP pathway (no MVA)
35163971	IspH reductive dehydroxylation / anti-infectives	Last enzyme of pathway; absent in mammals; oxygen-sensitive [4Fe-4S]
28862365	<i>E. coli</i> IspH:inhibitor crystal structures	Drug-target status; inhibitor-bound structures for rational design
24188825	Structure of <i>P. falciparum</i> HMBPP reductase	Identifies [2Fe-2S] ferredoxin as electron donor via docking
26118554	NRVS of <i>E. coli</i> LytB/IspH cluster	Apical iron linked to three waters in substrate-free enzyme; substrate/inhibitor complexes
40432238	Fe–S enzymes of MEP pathway: IspG and IspH (review)	Oxygen sensitivity and protective mechanisms of the Fe–S active site
31756006	Enzymatic evidence for rotation in LytB/IspH reaction	Independent evidence for the rotation/organometallic step in catalysis

Plant/orthologue corroboration. Functional studies of Arabidopsis HDR (the plant IspH orthologue) using site-directed mutagenesis confirmed that the conserved cysteines form the Fe–S cluster and identified substrate-binding/catalytic residues, with most residues "*absolutely conserved from bacteria to plants*" (PMID: 25037211, PMID: 25723575). These cross-kingdom conservation data reinforce the transferability of mechanistic conclusions to the *P. putida* enzyme.

Structural/computational support specific to Q88Q89. The AlphaFold model (mean pLDDT 96.2; Finding 6) and its 0.31 Å superposition on the substrate-bound *E. coli* structure (Finding 7) provide direct, protein-specific structural evidence that Q88Q89 possesses an intact IspH catalytic scaffold.

Limitations and Knowledge Gaps

- 1. No direct biochemical characterization of the *P. putida* protein.** All enzymatic, spectroscopic and crystallographic mechanistic data derive from orthologues (*E. coli*, *A. aeolicus*, *P. falciparum*, plants). The function of Q88Q89 is assigned by very strong homology/synteny/structure rather than by an in vitro assay of the purified KT2440 enzyme. Kinetic parameters (k_{cat}, K_m for HMBPP), the exact IPP:DMAPP ratio, and the identity/affinity of its cognate ferredoxin remain to be measured for *P. putida*.
- 2. Essentiality inferred, not directly demonstrated in KT2440.** Essentiality is strongly expected (sole precursor route; no MVA bypass) but was not confirmed here by a *P. putida* *ispH* knockout/conditional-depletion experiment. Available KT2440 Tn-seq/transposon fitness datasets (e.g.,

 40302248

 33964456

 32826213

were not specifically mined for *ispH* fitness scores in this investigation.

- 3. Electron-donor identity.** The physiological electron donor is annotated as reduced [2Fe-2S] ferredoxin (UniProt/RHEA; docking in *P. falciparum*), but the specific ferredoxin(s)/flavodoxin(s) and the ferredoxin-reduction system supplying IspH in *P. putida* have not been experimentally identified.

4. **Structural model, not an experimental structure.** No experimental crystal or cryo-EM structure of the *P. putida* IspH exists; the AlphaFold model does not include the [4Fe-4S] cluster or bound substrate, so cluster geometry and substrate contacts are inferred from the *E. coli* template.
5. **Product-ratio determinants.** The ~4–6:1 IPP:DMAPP ratio is a bacterial family generalization; active-site residues that tune this ratio (e.g., the Phe effect noted in Ginkgo, [P 28139297](#)) were not examined for the specific *P. putida* residues.

Proposed Follow-up Experiments / Actions

1. **Heterologous expression and in vitro assay.** Clone PP_0606, express in *E. coli* under anaerobic/Fe-S-reconstitution conditions, and assay HMBPP-reductase activity (e.g., photometric/HPLC detection of IPP + DMAPP), measuring kcat, Km(HMBPP), the IPP:DMAPP product ratio, and O₂ sensitivity — providing direct, protein-specific confirmation of Finding 1.
2. **Cofactor characterization.** Use EPR/Mössbauer/UV-vis on the reconstituted enzyme to confirm the [4Fe-4S]²⁺ cluster and apical-iron water ligation directly for Q88Q89 (validating Finding 2 for this protein).
3. **Genetic essentiality test.** Attempt a clean *ispH* deletion in KT2440 with and without an inducible complementing copy (or CRISPRi knockdown) to formally demonstrate essentiality and characterize the depletion phenotype (growth arrest, loss of quinone/carotenoid/bactoprenol-dependent processes).
4. **Mine existing fitness data.** Extract the PP_0606 fitness/essentiality call from published KT2440 RB-TnSeq/Tn-seq datasets ([P 33964456](#), [P 32826213](#), [P 40302248](#)) as rapid, no-cost genetic evidence for essentiality.

5. **Site-directed mutagenesis.** Mutate Cys12/Cys96/Cys198 and the annotated proton-donor residue 126 and substrate-binding residues, then test activity/complementation to confirm their catalytic roles in the *P. putida* enzyme (analogous to the Arabidopsis HDR study,

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6. **Electron-donor identification.** Identify and reconstitute the physiological ferredoxin/flavodoxin + reductase system of *P. putida* that reduces IspH, and measure electron-transfer kinetics.

7. **Inhibitor susceptibility.** Test known IspH inhibitors (HMBPP analogues such as AMBPP/TMBPP, and cell-penetrant prodrug variants) against *P. putida* and its purified IspH to evaluate the enzyme's tractability as an antibacterial target in this organism.

Conclusion

IspH (Q88Q89 / *ispH* / PP_0606) in *Pseudomonas putida* KT2440 is unambiguously identified as **4-hydroxy-3-methylbut-2-enyl diphosphate reductase (HMBPP reductase, EC 1.17.7.4)**, the oxygen-sensitive [4Fe-4S] metalloenzyme that catalyzes the final, committed step of the MEP isoprenoid-biosynthesis pathway. It performs the $2e^-/2H^+$ reductive dehydroxylation of HMBPP to the universal C5 precursors IPP and DMAPP in the cytoplasm, using reduced [2Fe-2S] ferredoxin as electron donor and an apical-iron / organometallic mechanism. Because this pathway is the sole route to isoprenoid precursors in *P. putida* and is absent from humans, IspH is an essential, non-redundant enzyme and a validated antimicrobial drug target. The annotation is supported by seven converging lines of biochemical, spectroscopic, structural, sequence, synteny and organism-specific metabolic evidence; the principal remaining gap is the absence of direct biochemical and genetic characterization of the *P. putida* protein itself.

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